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Can we meet the Target? Status and future trends for fisheries sustainability

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We assess progress towards Aichi Biodiversity Target 6, which aims to achieve global fisheries sustainability by 2020. Current trends suggest that the proportion of fish stocks within safe ecological limits is likely to decline until 2020. While model projections show a considerable reduction in overexploited stocks by 2050 if climate change is not considered, there will be a substantial increase in the risk of overexploited fish stocks if climate change is taken into account. Overall, although there is progress toward rebuilding fisheries in some developed nations, this improvement is insufficient to meet the Aichi Target by 2020; there is a need for substantial changes to current fisheries policy and management if Target 6 is to be met.

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Introduction

The Strategic Plan for Biodiversity 2011–2020 was adopted by the Parties to the Convention on Biological Diversity (CBD) in 2010. It includes a set of 20 Targets (the Aichi Biodiversity Targets) which aim to reduce biodiversity loss, improve the status of biodiversity, and enhance biodiversity benefits and ecosystem services to society by 2020. Target 6 has the aim that 'By 2020 all fish and invertebrate stocks and aquatic plants are managed and harvested sustainably, legally and applying ecosystem based approaches, so that overfishing is avoided, recovery plans and measures are in place for all depleted species, fisheries have no significant adverse impacts on threatened species and vulnerable ecosystems and the impacts of fisheries on stocks, species and ecosystems are within safe ecological limits.'

With 3 years left to the Target deadline, it is timely to assess whether or not the Target is likely to be achieved by 2020. Understanding where we stand now and how this translates to potential future scenarios in the short (to 2020) and long term (2050) is important to inform policy makers about where actions are needed in order to achieve the Target. As such, this paper first assesses evidence about progress towards Target 6 by reviewing the current status and trends of marine fisheries sustainability worldwide. We focus on the impact of fisheries on marine biodiversity and habitats, and the factors that impact upon fisheries sustainability. The second part of this study involves scenario modelling for marine fisheries, in which we forecast biological outcomes of fisheries in the long term. While Target 6 covers a broad range of issues, we limit the scope of our analysis about future fisheries sustainability to commercially targeted fish species.

The main research questions we address are:

1. Are we on track to meet Target 6 by the 2020 deadline?
2. What are the short and long term implications of current trends for fisheries and marine biodiversity?
3. What needs to be done to achieve Target 6?

A suite of measurable indicators for monitoring progress towards Target 6 was compiled by the CBD [1]. These were grouped into broad categories covering trends in first, certified sustainable fisheries; second, proportion of depleted, target and bycatch species with recovery plans; third, population and extinction risk in target and bycatch species; fourth, fishing practices; fifth, proportion of fish stocks outside safe biological limits; and sixth, catch per unit effort.

Methods

Literature review

We conducted a review of published and grey literature to assess the current status and trends of marine fisheries as they relate to achieving Aichi Target 6. As this is a global review, the studies covered both regional and global scales, thereby minimising regional biases. However, it

Table 1

Characterisation of Rio + 20 Pathways and how they will achieve 2050 target

Pathway	Main assumption
Decentralised Solutions	Focuses on decentralised solutions such as local energy production, agriculture that makes use of natural corridors, and national policies that regulate equitable access to food.
Global Technology	Focuses on large-scale technologically optimal solutions and a high level of international coordination, for example, intensive agriculture and trade liberalisation.
Consumption Change	Focuses on changes in human consumption patterns by limiting per capita meat intake and choosing less energy-intensive lifestyles.

Source: van Vuuren *et al.* [1]

is noted that literature on certain topics, such as fish stock assessments, are predominantly based upon analyses conducted in developed countries. The scope of this review included impacts of marine capture fisheries on marine biodiversity and ecosystems (e.g. bycatch, habitat modification), climate change impacts on fisheries, and projections for different indicators and aspects of fisheries sustainability in the short and long term. We also identified policy or management approaches that can elucidate what governments can do to move towards sustainable fisheries. The temporal scope of this review was from 2010 onwards in order to coincide with the Strategic Plan for Biodiversity timeframe, which covers the period 2011–2020. However, earlier literature was used if no recent relevant studies were available.

Projecting fisheries trends to 2050

Scenarios

To investigate fisheries sustainability trends to 2050, we projected the proportion of fish stocks at risk of overexploitation in 2050. This projection incorporated climate change impacts on fish biomass and distribution in 2050, and are based on the underlying assumptions of the three Rio + 20 Pathways [2] (Table 1), a set of alternative socio-economic pathways along which the sustainable development goals set out in the 1992 Rio Declaration (eradicating poverty, halting climate change, and conserving ecosystems) may be achieved. Each of the three Rio + 20 Pathways (Decentralised Solutions, Global Technology, Consumption Change) describe the level of effort and changes in governance and socio-economic policies required to achieve these sustainable development goals by 2050.

The Rio + 20 Pathways tend to focus mainly on terrestrial sectors such as forestry, energy, and agriculture; we applied the same concept of the Rio + 20 Pathways to the marine realm. We adapted the governance and socio-economic conditions of the three Rio + 20 Pathways to potential scenarios for how sustainable marine fisheries (i.e. fisheries exploited at maximum sustainable yield (MSY)) may be achieved by 2050 (Table 2, with details in Supplementary Materials). Assumed changes in fishing effort for coastal and high seas fisheries, and governance and policy adjustments are consistent with the broader societal development trajectory in each of the Rio + 20 Pathways. Although not stated in Target 6, we use MSY as a reference point for sustainable fisheries because it is widely accepted as an objective for fisheries management, and is the measure used in international agreements and forums such as the United Nations Convention on the Law of the Sea[1] and the 2002 World Summit of Sustainable Development [3]. It is also used by various inter-governmental institutions such as the Food and Agriculture Organization of the United Nations and European Union Common Fisheries Policy. It is noted that fisheries can still be productive at levels below MSY [4].

Projecting future fisheries trends

Current global marine catch is around 81.5 million t [5], while estimated potential catch at maximum sustainable yield (MSY) is around 88.7 million t, ranging from 82.7 to 99.4 million t [6]. Therefore, global fisheries are currently at about 90% of MSY level. To investigate future fisheries trends (i.e. proportion of fish stocks at risk of overfishing), we used global fisheries catch data from the *Sea Around Us* project (www.seaaroundus.org) with a population dynamics model developed by Martell and Froese [7]. We defined fishery stocks by species and FAO (Food and Agriculture Organization of the United Nations) statistical area. The global ocean is divided into 27 major fishing areas for statistical purposes by the FAO. We only included stocks with catch data reported at the species level. Based on the combinations of fish stocks found within all FAO Areas from the *Sea Around Us* project database (www.seaaroundus.org), we obtained a total of 1343 stocks within Exclusive Economic Zones (EEZs) and 537 stocks in the high seas, inclusive of fishes and invertebrates.

We applied the Catch-MSY method [7] to simulate changes in fish stock biomass and exploitation rate. The Catch-MSY method is a biomass dynamic model that is run with time-series of catch removal (based on catch data) and a Monte-Carlo simulation (N = 100 000) of

[1] United Nations Convention on the Law of the Sea 10 December 1982 Overview and full text. http://www.un.org/Depts/los/convention_agreements/convention_overview_convention.htm.

Table 2

Rio + 20 Pathways adapted to scenarios for marine fisheries. See Supplementary Material for full narrative

Potential future	Rio + 20 Pathway		
	Dentralised Solutions	Global Technology	Consumption Change
Direction/focus area for fisheries development and policy	-Improved governance through use of local and participatory solutions. -Reduction in capacity enhancing subsidies	-Intensive production (aquaculture of piscivores), leading to subsequent increase in demand for forage fish	-Expected increase in demand for fish based diet -Use of 'greener' fishing technology - Reduction in bycatch - Aquaculture of herbivores
Predicted trends in catch and fishing effort	-High seas catch↓ -Coastal catch ↑ -Fishing effort at -F_{MSY} level by 2050	-High seas and coastal catch remain at status quo at 2050 -Coastal small pelagic purse seine effort ↑ by 2% per year until 2020 then remains constant to 2050	-F_{MSY} reached for all species by 2020. -Targeted high seas stocks are rebuilt by 2050. -High seas catch↓ by 2050. -Coastal catch maintained at current level at 2050 -Coastal non-tuna purse seine ↓ by 1% per year to 2020 then remains constant to 2050
Exploitation rate in 2050	- Overexploited populations have recovered and sustainable catch levels are attained by 2050. Therefore there are no overexploited stocks, and both high seas and coastal fisheries are at 100% MSY level	-Both high seas and coastal fisheries catches remain at current exploitation status (90% MSY)	-High seas stocks are rebuilt but coastal fisheries remain overexploited -High seas fisheries are at MSY level in 2050, while coastal fisheries remain at the current level (90% MSY)

stock biomass with random values of the intrinsic population growth rate (r) and carrying capacity (K). The ranges of r and K values from which the Monte-Carlo simulations were drawn were based on the 'resilience' categorization of FishBase (www.fishbase.org) and the maximum catch of the time series (see Martell and Froese [7] for details). Only those simulation runs that generated reasonable results were accepted ('reasonable results' were defined as having never collapsed the stock or exceeded carrying capacity, and the resulting biomass falling within the assumed range of depletion) [7].

The simulation time-frame was from 1950 to 2006. In addition, exploitation rate (E_t) was predicted for each simulation run from simulated annual biomass (B_t) and catch (C_t) at year (t) where $E \approx C/B$. We also estimated the maximum sustainable yield ($MSY = K * r/4$) of each major fish stock and the fishing effort required to achieve MSY ($r/2$).

Determining fish stocks at risk of over-exploitation

Based on the simulation results from the Catch-MSY model, we calculated the proportion of stocks that have a high risk of overexploitation for each FAO area. We used a threshold of 40% of initial carrying capacity (K) without climate change effect as a criterion to define overexploitation — when stock biomass is below 0.4 K, we considered the stock to be overexploited. This is consistent with the precautionary approach used by Canada's Department of Fisheries and Oceans, which sets 0.4 K as an upper reference point, that is, stocks that fall below this point are considered to be in a cautious or critical zone [8]. Similarly, the US Pacific Fisheries Management Council defines target biomass as 40% of unfished biomass (0.4 K) for groundfish species [9].

For each stock, we counted the number of accepted runs generated by the Monte-Carlo simulation that were overexploited. If it was more often than not (i.e. probability >50%, binomial test $P < 0.05$) that the stock would be overexploited, we considered the stock to have high risk of overexploitation. This was calculated for each stock in year 2006. We then used the accepted simulations from the Catch-MSY model to project future changes in stock size from 2007 to 2050 under different scenarios of fishing and climate change. Fishing scenarios defined by the Rio + 20 Pathways were incorporated into the model through assumed changes in exploitation rate over time. Climate change and fishing scenarios are detailed below.

Future climate and fishing scenarios

Climate change is expected to affect the distribution of fish through changes in habitat suitability [10]. Hence, we projected future changes in habitat suitability under climate change for each fish species using three different species distribution models: MAXENT, Aquamap and Dynamic Bioclimate Envelope Model [10,11]. The climate change scenario used was the Representative Concentration Pathway (RCP) 8.5 scenario [12] that is projected to lead to a 4 °C increase in average global surface temperature by 2100 [13].

We calculated the projected changes in average habitat suitability for each stock for each year from 2007 to 2050. The estimated changes in habitat suitability were then used to drive the Catch-MSY model for future projections by assuming that changes in carrying capacity (K) of each stock in the future is directly proportional to changes in habitat suitability. We assumed that fisheries management does not adapt to climate change to illustrate the likely outcomes under status quo conditions (i.e. no additional efforts to move towards Target 6). Therefore, reference points of biomass and exploitation rate at MSY remain constant over time for each stock under climate change.

Results

Assessment of evidence about progress towards Aichi Target 6

Status and trends

Recent studies agree that marine fisheries are, in general, overexploited, although there is disagreement on the extent of overexploitation, and the status and trends of global marine fisheries [14]. One study showed that 63% of 166 assessed fish stocks (the majority of which were well managed, developed country fisheries) have lower biomass levels than required to obtain maximum sustainable yield (MSY) [15]. While rebuilding had not yet led to overall biomass recovery, these assessed stocks were found to have the potential to recover where low exploitation rates were maintained [15]. This has since been demonstrated for the Northeast Atlantic, where exploitation of the major fish stocks has declined significantly during the last decade and stock biomasses are rebounding [16]. Another study provided a more positive picture, stating that the proportion of fished stocks that are overexploited or collapsed has remained stable in recent years, and that rebuilding efforts for these fisheries have reduced exploitation rates [17].

At the global level, catch trend analysis shows a less optimistic situation compared to stock assessments [18,19]. According to the FAO, 57% of assessed marine fish stocks are considered fully exploited (i.e. at or near the maximum sustainable yield), 30% are overexploited, and the remaining 13% are non fully-exploited [20]. The percentage of overexploited stocks has remained in the 25–30% range for the past 20 years. The most recent State of the World Fisheries and Aquaculture report (SOFIA) indicates that in 2013, 31% of assessed marine fish stocks were considered to be fished at a biologically unsustainable level. Fully fished stocks accounted for 58% of assessed stock, while underfished stocks accounted for 11% [5].

Unlike the trend from stock assessments, the continuous declining trend from catch data does not stabilise [18,21••]. Rather, the percentage of non fully-exploited (under and moderately exploited) stocks has decreased gradually through time, whereas the percentage of overexploited and depleted stocks has increased [18]. In a recent study of over 1793 unassessed fisheries, it was found that 64% of these fisheries had lower stock biomass than required to support MSY, and that 18% of unassessed stocks were collapsed [22]. While all unassessed stocks were on a declining trend, 64% of these stocks could potentially increase sustainable harvest if they were rebuilt.

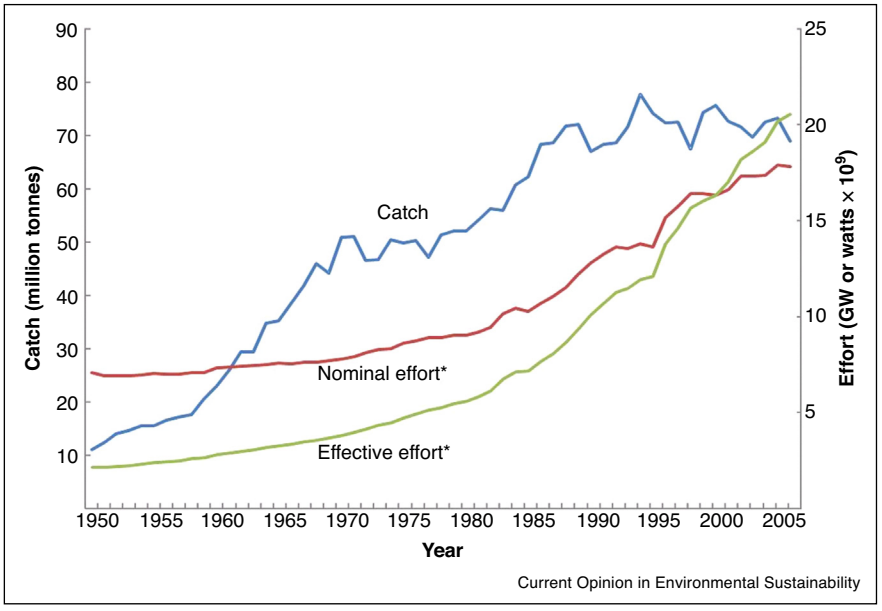
An analysis of 8 indicators of fishing pressure, state, benefits, and responses of fisheries indicated an overall decline in global marine fisheries and long-term fisheries benefits [23]. Furthermore, despite management and policy actions taken by coastal states, pressures on fisheries are increasing [23]. In 2000 alone, overfishing resulted in potential catch losses that amounted to 7–36% of actual landed tonnage that year [6], although a subsequent re-estimation showed that this range was actually low [24]. Analysis of catch and primary production data also showed an increasing trend of ecosystem overfishing (i.e. overfishing that leads to an alteration in ecosystem diversity, productivity, variability, and species composition) from 1950 to 2000 [25].

There has been a global expansion of marine fisheries over the last 60 years that fisheries have been monitored [26]. Similarly, the number of fishers has increased from 1970 to 2010 [23]. Fishing effort measured in total kilowatt days shows an increasing trend; nominal effort more than doubled from 1950 to 2010, suggesting a global decline in catch per unit effort (Figure 1) [27]. The use of bottom trawls, which can directly harm benthic habitats and shift the benthic composition towards small opportunistic species, has increased globally in marine ecosystems [28] (Figure 2).

The persistence of overfishing in the world's oceans will continue to negatively affect marine biodiversity and ecosystems [29,30]. In certain cases, fishing has resulted in collapse and local extinction of marine species [31,32]. Currently, over 550 species of marine fishes and invertebrates are listed on the IUCN Red List as Critically Endangered, Endangered, and Vulnerable. This may be an underestimate in itself, due to insufficient data to assess the conservation status of many marine organisms. In particular, many deep sea fishes and other large bodied, slow growing fishes are especially vulnerable to over exploitation [33,34]. 'Fishing down marine food webs' occurs when higher trophic level fish are progressively depleted, and replaced with lower trophic fish — a process that has been documented in many ecosystems [35,36] albeit debated [37].

The impact of fisheries on biodiversity is further exacerbated by factors that directly affect fish populations, the physical marine environment, and ecosystems. Marine

Figure 1



Global trends in estimated fisheries catch and fishing effort (nominal and effective) 1950–2006.
Source: Watson *et al.* [27].

pollution has caused changes in the structure and function of phytoplankton, zooplankton, benthic and fish communities, and also caused immune system responses and interruptions in the life cycle and physiological development of marine organisms [38,39]. Land reclamation, eutrophication, disease, and direct exploitation has led to the loss of wetlands, seagrasses, and other submerged aquatic vegetation [40]. It appears that 41% of 20 assessed marine ecosystems worldwide are strongly affected by multiple anthropogenic drivers, and no one area is unaffected by human activities [41]. The biological and ecological impacts of fishing also affect fisheries participants. In particular, the higher catching power of industrial scale fisheries has decreased inshore resources and increased competition for increasingly scarce resources throughout the world. This has negatively affected the societal well-being of small-scale artisanal communities worldwide, including their livelihoods, subsistence economies, and culture [42–45].

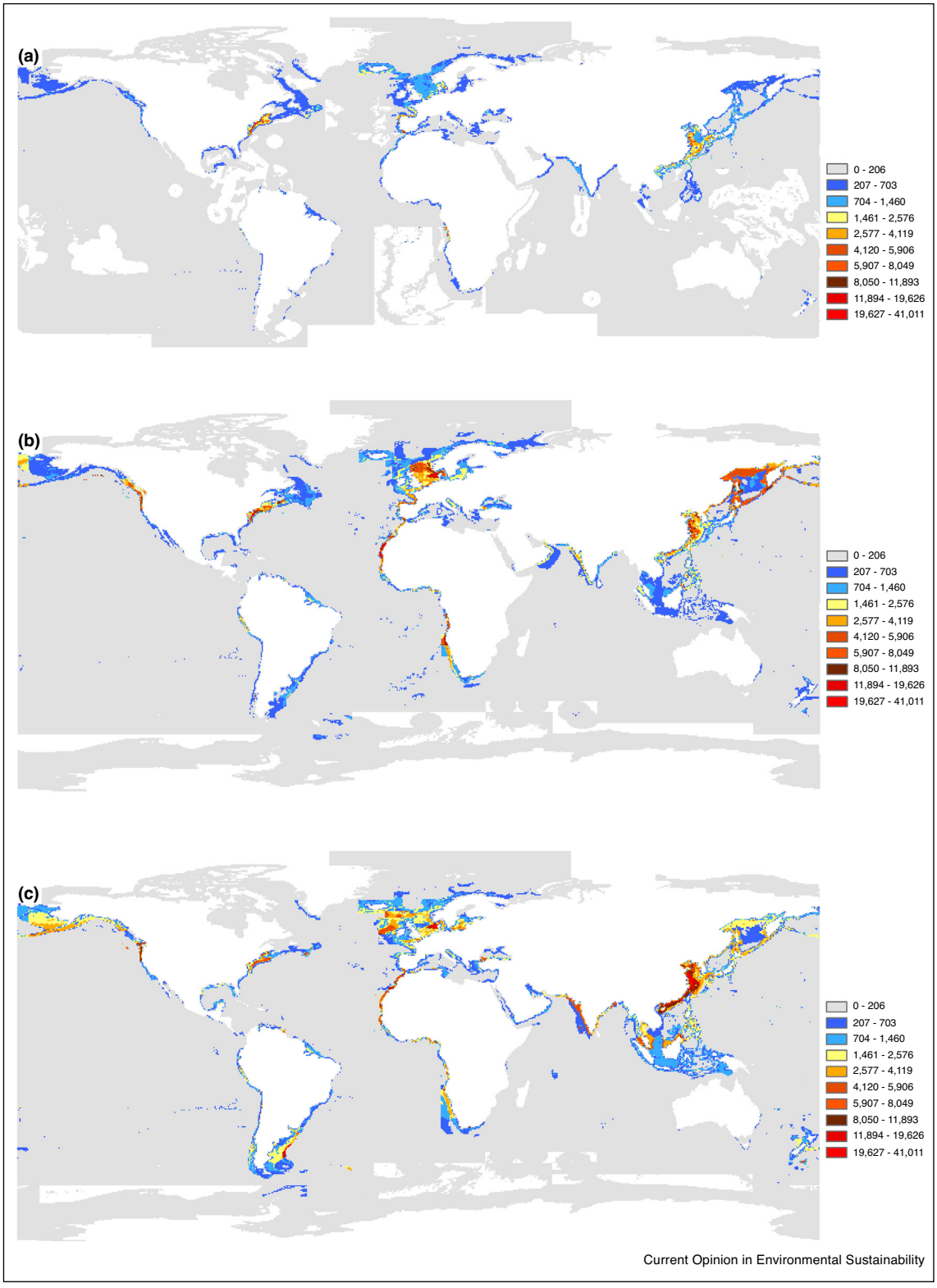
A positive trend is that the number of countries ratifying the UNCLOS (United Nations Convention on the Law of the Seas) increased annually since 1982, reaching 161 countries in 2010 [23]. However, an assessment of 53 countries that landed 95% of world fish catch showed that their overall compliance with the FAO Code of Conduct for Responsible Fisheries was low, with over 60% of countries failing and none obtaining an overall 'good' grade [46]. Although the FAO Code of Conduct is voluntary and states are encouraged to apply the codes they deem relevant, the overall poor performance demonstrates a low priority placed on fisheries management.

Similarly, an evaluation of ecosystem based fisheries management found that out of 33 countries that landed 90% of world fish catch, over half failed, none received a 'good' rating, and only 4 were 'adequate' [47]. Further, a global assessment of overall management effectiveness found that only 5% of all Exclusive Economic Zones (EEZs) were in the top quarter of the scoring scale [48]. Importantly, high income EEZs had significantly better overall management than low income EEZs. Factors that contributed to low management effectiveness in high income EEZs were subsidies and excess fishing capacity, whereas deficient scientific, political, and enforcement capacity contributed to low effectiveness in low income EEZs [48]. On a more positive note, there are examples of successful fisheries management in rebuilding fish stocks [49]. In addition, implementation of co-management models at the community level was found to be associated with successful fisheries [50].

Projecting forward to 2020

Demand for fish is expected to grow [51,52••], as fish and fishery products will continue to be highly traded, with 36% of world fish production projected to be exported in 2022 [53]. Combined with the continued spread of human impacts from the coastal zone to deep sea, it is expected that past trajectories of biodiversity loss and reduced ecosystem resilience will forecast future changes in the ocean if no measures are put in place to stop the trend [54••].

Figure 2



Global expansion of bottom trawling. Maps show global distribution of catches from trawling at different time periods **(a)** 1950–1960; **(b)** 1970–1980; **(c)** post-2000 (units are tonnes of catch). Based on the database of Watson *et al.* [27].

It is possible for marine ecosystems to recover if exploitation rates are substantially reduced. Despite the extended time needed for marine species and ecosystems to recover (e.g. fish stock recovery requires 4–26 years, while ecosystem recovery ranges from 10 to 42 years [55]), recent progress has been made in well developed and managed fisheries in North America, New Zealand, and Europe where current exploitation rates are predicted to achieve a conservation target of less than 10% collapsed stocks [15,17]. Therefore, although it is unlikely that all overexploited stocks were restored to a level that produced MSY by 2015 [20,23], rebuilding policies and legislation can still potentially shift the trend towards achieving Target 6 in certain fisheries by 2020.

It is unlikely that global fisheries catch will change significantly in the next 20–30 years [51,56] unless substantial improvements in fisheries policy occurs. Projected trends of effective trawling effort show an increase to 2020, while the proportion of fish stocks that are within safe biological limits is projected to decline in the same period [57]. Overall, notwithstanding several positive rebuilding results in developed country fisheries, the projection forward to 2020 will most likely reflect past trends — that is, increasing exploitation rates in most of the world's fisheries [58••] (except for several developed countries and where market drivers make it uneconomic to fish), accompanied by declining catch rates and biomass of exploited species. Moreover, there is an urgent need to drastically reduce exploitation rates of vulnerable marine animals [59••]. Thus, despite divergent views about the current status of global fisheries, it appears that having all fish stocks that are exploited at, or rebuilt to a safe biological level (defined conceptually as having a biomass that is above biomass-at-maximum sustainable yield) by 2020 is unlikely, unless attaining the MSY objective is relaxed [4].

Scenario projections

Fisheries scenario projections to 2050

Based on the Catch-MSY model, the proportion of stocks at risk of overexploitation (i.e. those with greater than 50% probability of being overfished) was estimated for each FAO area in 2006 and 2050. Under current conditions, the Northwest Atlantic had the largest proportion (80%) of EEZ stocks that was predicted to be at risk of being overexploited, while the Arctic Sea had the lowest proportion (36%). For high seas fisheries, the Northeast Pacific and Arctic Sea had the highest and lowest proportion of stocks predicted to be at risk of being overexploited, at 88% and 25%, respectively (Table 3).

The status quo picture changes for 2050 projections under each Rio + 20 scenario, as outlined below and summarised in Table 4:

Decentralised solutions: Overall, this scenario resulted in fewer stocks at risk of overexploitation in both EEZ and high seas relative to current conditions. The Western Central Pacific and Southeast Atlantic (SW Africa) had the lowest and highest proportion of stocks at risk of overexploitation in EEZs, respectively. The Southwest Pacific (SE Australia and New Zealand) had the highest proportion of at risk stocks in the high seas, while none were projected for the high seas of the Arctic Sea.

Global technology: Among the 3 scenarios, Global Technology projected the highest proportion of stocks at risk of overexploitation in EEZs. In general, the proportion of EEZ stocks at risk of overexploitation was lower than the status quo, except in the Arctic Sea and Eastern Indian Ocean; the Eastern Indian Ocean had the highest proportion of at risk stocks, while the lowest proportion was projected in the Northwest Pacific. Similar to current conditions, the Arctic Sea and Northeast Atlantic had the lowest and highest proportion of high seas stocks at risk of overexploitation, respectively.

Consumption change: The proportion of at risk stocks in both EEZs and high seas were lower relative to current conditions, except for EEZ stocks in the Arctic Sea. In EEZs, the Northwest Pacific had the lowest proportion of stocks at risk of overexploitation, while the Eastern Indian Ocean had the highest proportion, followed closely by the Eastern Central Atlantic (W Africa), Southeast Atlantic (SW Africa), and Southwest Pacific (SE Australia & NZ). As with the other two scenarios, the lowest and highest proportion of high seas stocks predicted to be overexploited occurred in the Arctic Sea and Northeast Atlantic, respectively.

Without consideration of climate change, all scenarios resulted in a considerable reduction in at risk stocks by 2050 in both EEZs and high seas (Table 4). This is because in all scenarios, exploitation rates for most species are assumed to be set at a sustainable level (i.e. at a level required to achieve MSY). The proportion of stocks at risk of being overexploited was higher in the Global Technology and Consumption Change scenarios because fishing for small pelagic species was assumed to be intensified in these two scenarios. Further, under both these scenarios the eastern Indian Ocean was projected to have the highest proportion of at risk stocks. There are residual risks of overexploitation even under the Decentralized Solution scenario, which emphasizes improved local and participatory governance, because a small proportion of slow growth and low productivity stocks that are currently overexploited may take more than 40 years to fully recover if drastic measures such as full fishery closures are not implemented.

Under the high climate change scenario (high emissions, RCP 8.5), the model projected substantial increase in risk

Table 3

The proportion of stocks that are overexploited (i.e. probability of overfishing >50%) in each FAO region in 2006

FAO Area	FAO Area name	Proportion of overexploited stocks	
		EEZ	High Seas
18	Arctic Sea	0.36	0.25
21	Atlantic NW	0.80	0.75
27	Atlantic NE	0.59	0.86
31	Atlantic WC	0.62	0.60
34	Atlantic EC	0.66	0.79
37	Mediterranean and Black Sea	0.49	–
41	Atlantic SW	0.55	0.66
47	Atlantic SE	0.78	0.73
51	Indian Ocean W	0.49	0.33
57	Indian Ocean E	0.40	0.43
61	Pacific NW	0.71	0.74
67	Pacific NE	0.68	0.88
71	Pacific WC	0.47	0.32
77	Pacific EC	0.85	0.77
81	Pacific SW	0.62	0.51
87	Pacific SE	0.73	0.74

NW = northwest, NE = northeast, WC = western central, EC = eastern central.

Table 4

The proportion of stocks with probability of overfishing >50% by 2050 under the three Rio + 20 Pathway scenarios. The values are reported as EEZ/High Seas

FAO Area	FAO Area name	Proportion of over-fished stocks		
		Decentralised Solutions	Global Technology	Consumption Change
18	Arctic Sea	0.29/0.00	0.50/0.00	0.43/0.00
21	Atlantic NW	0.32/0.33	0.46/0.33	0.44/0.33
27	Atlantic NE	0.31/0.42	0.43/0.42	0.39/0.42
31	Atlantic WC	0.19/0.28	0.44/0.28	0.42/0.28
34	Atlantic EC	0.30/0.33	0.50/0.33	0.49/0.33
37	Mediterranean and Black Sea	0.24/–	0.41/–	0.39/–
41	Atlantic SW	0.26/0.29	0.48/0.29	0.46/0.29
47	Atlantic SE	0.33/0.24	0.51/0.24	0.49/0.24
51	Indian Ocean W	0.16/0.10	0.48/0.10	0.48/0.10
57	Indian Ocean E	0.14/0.30	0.55/0.30	0.50/0.30
61	Pacific NW	0.13/0.26	0.35/0.26	0.30/0.26
67	Pacific NE	0.27/0.25	0.49/0.25	0.37/0.25
71	Pacific WC	0.11/0.16	0.52/0.16	0.40/0.16
81	Pacific EC	0.29/0.33	0.47/0.33	0.43/0.33
87	Pacific SW	0.29/0.41	0.51/0.41	0.49/0.41
Global (no CC)		0.24/0.30	0.31/0.30	0.28/0.3
Global (RCP 8.5)		0.42/0.67	0.49/0.67	0.46/0.67

of overexploitation for both EEZ and high seas stocks. For the Decentralized Solution scenario, stocks at risk of overexploitation almost doubled under climate change (RCP 8.5), while the increase in fishing effort for pelagic fishes under the Global Technology scenario further increased the at risk stocks. Although all scenarios resulted in a decrease in stocks at risk of being overexploited relative to the current level, climate change is expected to substantially increase the risk of not achieving Target 6.

Discussion and conclusion

We find that given present status and trends, global fisheries are not likely on track to meet Target 6 by the 2020 deadline. While some indicators show variable progress, (e.g. recovery plans and measures in place, number of certified sustainable fisheries), others do not. In particular, despite some regional progress, trends in the proportion of fish stocks outside safe biological limits, catch per unit effort, and destructive and IUU fishing do not show overall improvement [57]. This is

supported by modelling results which indicate that fish stocks are at risk of being overexploited in most of the world's fishing regions.

If Target 6 is not achieved, trends in marine species loss and degradation of marine ecosystems will likely continue, resulting in: first, decline in abundance of targeted and non-targeted species; second, loss of keystone species and top predators from marine ecosystems, potentially causing shifts to alternate ecosystem states (e.g. coral–algal systems); third, unsustainable levels of bycatch and discards, including vulnerable species of marine megafauna; fourth, decrease in marine trophic index, impacting fish community structure and food webs; and fifth, loss or degradation of marine habitats.

If Target 6 is achieved, there is potential for depleted marine species and ecosystems to recover [55,60]. Rebuilding efforts has resulted in the recovery of some fish populations in well managed fisheries in developed and developing countries [15,16], although recovery may not be common among all fish and marine animals [30,55]. It is suggested that recovery only occurs in 10–50% of species or ecosystems, thus indicating the need for much improved management and conservation [55]. Ecosystem shifts to an alternate state and long-term changes in environmental conditions add to the uncertainties on the possibility and rate of recovery.

In light of future climate variability and change, a less heavily fished marine system is likely to produce more stable catch than an overexploited one [61]. As such, failure to meet Target 6 will likely make fisheries management more challenging because of the direct and indirect impacts climate change will have on marine biodiversity and fisheries catch through effects on the physiology and behaviour of fish, as well as through changes in the physical and chemical properties of the marine environment [62••,63]. For instance, maximum body size and growth of fishes are projected to decrease by 14–24% by 2050 (relative to current time) [64]. Changes in species composition of fisheries catch has been partly attributed to long-term ocean temperature changes [65], which is causing commercially important species to shift their distributions, mainly towards higher latitude and deeper waters [66,67]. Overall, it is expected that fish diversity, productivity, and yields will increase at high latitudes and decrease at mid to low latitudes [68,69].

The expected poleward movement of species may also crowd out endemic species [70], change the composition of fish assemblages [71], and increase the risk to critically endangered species such as the common skate and angel shark in certain areas [11]. Climate change is expected to affect fish population dynamics through effects on recruitment [61] and alterations to marine food web structure [72], while changes in primary production is expected to affect marine food webs and ecosystems [70]. Future net primary production changes predicted by 2050 may increase marine fisheries for 'large' fish species by 6% in 69 EEZs, while those for 'small' fish increase by 3.6% in the top fishmeal producing countries [73]. In addition, ocean acidification and deoxygenation are expected to reduce habitats for exploited marine organisms and fisheries yield in some regions [66]. These potential climate induced changes to fisheries [74••], and their linked societal effects, reinforce the need for attaining sustainable fisheries that have the capacity to respond and adapt to evolving conditions.

It is possible for marine ecosystems to sustain per capita fish consumption rates through to 2050 if effective fisheries management policies are implemented and technological improvements are made [73]. To move towards Target 6, current excess fishing capacity has to be drastically reduced. This encompasses eliminating or diverting harmful subsidies [75] and stopping Illegal, Unreported, and Unregulated (IUU) fishing, which is estimated to take at least 35% of global catches [76], and has led to depletions of certain fish stocks [77].[2] Eliminating destructive fishing gears that damage vulnerable marine habitats (especially rapidly declining coral reefs, seagrasses, and cold water coral and sponge grounds [78]) and have high bycatch is essential for minimising biodiversity and ecosystem impacts. Bycatch, which amounts to about 40% of annual global marine catch [79], is still not managed adequately at the regional level [80] despite a recent trend towards reduced discards [21••,80]. Unselective fishing is also considered a primary driver of population declines in some species of marine megafauna [81,82], with around 600 000 marine mammals caught globally as bycatch every year [83].

Common management tools used to reduce exploitation rates include gear restrictions, creating marine protected areas, and the use of economic incentives (e.g. vessel buybacks, individual transferable quotas (ITQs)) to encourage reduced levels of fishing effort [15,84]. Social and economic assistance programmes that provide retraining and business or financial assistance have been used in some countries such as Canada, Norway, and Australia, and are important for helping displaced fishers transition to other employment [85].

Fisheries regulations have to be viewed as legitimate by stakeholders in order to gain their support and compliance. Devolution of governance to indigenous peoples and local communities, shared governance, and co-management arrangements have contributed to successful fisheries management outcomes [50], especially in small-scale fisheries in developing countries [86••]. This

[2] It is also possible that in some cases the direction of causality is that excess capacity may lead to IUU fishing.

situation is consistent with the Decentralised Solutions scenario, which our projections showed resulted in the lowest proportion of overexploited stocks by 2050. Given that the majority of the world's fishers are engaged in small-scale fishing, the use of shared governance and co-management arrangements is a promising action that can lead towards sustainable fisheries, bearing in mind that co-management can also overlook crucial dimensions of governance [87], and lead to undesirable social and ecological outcomes [88]. Instead of a single management system, some authors recommend that ecosystem-based management approaches perform best [14]. Market-based mechanisms such as certification also have considerable potential to be effective tools [89,90].

It should be noted that progress towards sustainable fisheries management is mainly being made in European and North American fisheries. For instance, as of 2013, 21 of 44 fish stocks requiring rebuilding in the United States were considered to be rebuilt, while 7 had made significant rebuilding progress [49]. The European Union (EU) Common Fisheries Policy, which regulates all fishing activities in European waters, improved the status for commercially exploited stocks in the North East Atlantic, North Sea, and Baltic Sea [91], although others have been less positive about the Policy's effectiveness [92]. In 2013, the EU reformed the Common Fisheries Policy in order to move towards sustainable management of all fish stocks by 2020.

In order to progress towards Target 6 worldwide, fisheries managers need to put more emphasis on being flexible and taking a precautionary approach to prevent overfishing, rather than reacting to it [93•], especially given the added uncertainties of climate change impacts on marine ecosystems and species. Indeed, the modelled scenario projections suggested that climate change substantially increased the proportion of stocks at risk of overexploitation for all scenarios. Feasible alternative livelihood or income options should also be considered, particularly for developing country small-scale fisheries, where it has been argued that management interventions based on rent or wealth based models are inappropriate; policies should instead invest in areas such as fishers' health and education, governance improvement, and addressing justice, social security, and human rights [94,95].

Our modelling approach was simplistic by necessity due to the number of stocks we were assessing. We acknowledge that this may have resulted in the categorization of some sustainably fished stocks as being overfished, as the model predicted a substantially higher proportion of overexploited stocks in some regions than the 30% assessed by the FAO for stocks worldwide [96]. Using 0.4K as the threshold to define overexploited stocks is consistent with a precautionary approach; higher values of K may have resulted in a less pessimistic outlook about the global status of fish stocks. Therefore, our projections may be considered a cautionary scenario, but does not change the urgency for policies measures at regional and international levels to prevent fish stocks from falling to the modelled levels.

The projections to 2050 used a single-species biomass dynamic model, which does not consider multi-species interaction through fishing gears or food-web dynamics [29]. In reality, MSY may be different to reach for each species concurrently because of factors such as bycatch and predator–prey interactions. Fisheries management was assumed to be non-adaptive to changes in stock abundance and productivity under climate change. While this may be valid for a short time frame, reference points for fisheries management are likely to be updated to changing ecosystem conditions. This highlights that adaptation to change, including climate change, is needed if Target 6 is to be met.

The combined climate change and fishing exploitation scenarios suggest that the same policy scenarios will have substantially different regional outcomes. Therefore, rebuilding strategies have to take into the account the prevailing socio-political context in identifying feasible management options [97], as well as be adapted to regional fisheries context and management frameworks. We have not assessed the cost of actions associated with each Rio + 20 scenario used in our model, but note that recent research indicates that rebuilding makes economic sense in the long term [98]. This is especially pertinent for priority areas such as the Western Central Pacific, where climate induced changes are predicted to have a pronounced impact, but where coastal populations tend to be most reliant on fisheries resources for livelihood and food security [99].

In conclusion, the current trajectory of fisheries strongly suggests that overall, global progress is insufficient to meet Target 6 by 2020, although it may be achieved regionally in select fisheries. Sustainable fisheries and recovery of marine biodiversity continue to be overwhelmed by local and global threats; consequently, significant changes in policy and practices are urgently required if Target 6 is to be met. In order to be effective, the challenge is for these policies to expand the bounds of conventional fisheries management to encompass addressing the human and institutional incentives driving fisheries exploitation at local and regional scales.

Conflict of interest

The authors report no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at https://doi.org/10.1016/j.cosust.2018.02.006.

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